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coding efficiency, but do not have the ability to trace the source. The compressed video information cannot be directly used for content traceability. In related technologies, for example, Chinese patent document CN115914677A provides an intelligent video security networking device and server. The device includes: a video gateway module and an intelligent analysis module. The video gateway module includes: a video access unit for receiving front-end image collection. The video stream data collected by the module; the video transcoding unit is used to decode the video stream data to obtain the original data stream; and transcode the video stream data to obtain video transcoding data; the intelligent analysis module is used to It is used to intelligently analyze the original data stream to obtain image coding and structured information; the video transcoding unit is also used to sign and encrypt video transcoded data, image coding and structured information; the video gateway module also includes: video forwarding unit For packaging signed encrypted files. However, this solution has at least the following technical problems: the intelligent analysis module only performs intelligent analysis on the original video stream to obtain image coding and structural information. This information is relatively limited and is not enough to fully describe the video content, such as the lack of scene information, target detection information, etc. Incomplete metadata will reduce the efficiency of subsequent traceability; due to the discontinuity in the insertion of metadata and watermark information, there will be gaps in the generated traceability information, which is not conducive to the reconstruction of a complete source-to-end communication link. Based on this, it is necessary to research a safe and traceable video encoding and decoding system to improve the security and confidentiality of video encoding and transmission, and to quickly complete video traceability after repeated video recording and dissemination events.

Contents of the invention

1. Technical problems to be solved In view of the problem of low video traceability efficiency existing in the prior art, the present invention provides a safe traceable video encoding and decoding system, by inserting traceable information into each link of video processing, and collecting and analyzing these at the display end. Information can effectively trace the source of video content, improve video traceability efficiency, and thereby improve security and confidentiality.

2. Technical solutions The object of the present invention is achieved through the following technical solutions. Embodiments of this specification provide a secure and traceable video encoding and decoding system, including: a video collection module, used to collect source video streams, and insert watermark information and metadata into the source video streams; a video encoding module, used to receive source videos stream, and inserts encoding node information into the source video stream, and compresses and encodes the video stream after inserting the encoding node information; the transmission module is used to encapsulate and transmit the encoded video stream output by the video encoding module; the decoding module uses Receive the video stream output by the transmission module, decode the received video stream, and insert decoding node information and watermarks into the decoded video stream; the display module is used to display the video stream output by the decoding module and parse the node information therefrom and watermarks to generate the transmission trajectory of the video stream; among them, the display module

contains a traceability analysis unit, which is used to obtain the decoded video stream, analyze the node information and watermarks in the video stream, and track the propagation link of the image. Further, the video acquisition module includes: a video source unit, used to obtain the source video signal generated by the video output device; a coding standard selection unit, used to select the corresponding video coding standard according to the format and system requirements of the source video signal. Video coding The standard includes H.264 and H.265; the video acquisition unit is used to collect the source video signal according to the selected video coding standard and generate the source video stream that meets the requirements of the video coding standard; the image enhancement unit is used to enhance the source video The stream is subjected to image quality enhancement processing, which includes denoising and color correction; a watermark information unit is used to insert watermark information at a preset position of the source video stream. The watermark information includes video collection time, location and device number information; metadata The generation unit is used to extract frame images in the source video stream as key frames at preset intervals, and generate metadata describing the video content by analyzing the key frames. The metadata contains target detection and scene recognition information; acquisition control The unit is used to control the working mode of the video collection module according to the network status and user instructions; wherein the video output device includes a camera and a computer operating system desktop. Further, the video encoding module includes: an information insertion unit, used to receive the source video stream, insert node information into the source video stream, and generate a video frame sequence with node information. The node information includes encoding time, location and encoder number. Information; encoding unit, used to receive a video frame sequence with node information, select the H.264 or H.265 video coding standard according to the network status, adjust the code rate control parameters and variable block size parameters of the encoder, and encode the video frame sequence Perform compression encoding and output the compressed and encoded video frame sequence; the encryption unit is used to receive the compressed video frame sequence, encrypt the video frame sequence using a symmetric or asymmetric encryption algorithm, and output the encrypted video frame sequence; the encapsulation unit , used to receive the encrypted video frame sequence and encapsulate it into an output code stream that complies with the video transmission protocol standard. Further, the transmission module includes: an encryption unit, used to receive the encoded video frame sequence output by the video encoding module, use an encryption algorithm to obfuscate it, and output the obfuscated video frame sequence; an encapsulation unit, connected to the encryption unit, to convert the The obfuscated video frame sequence is encapsulated into transmission data packets; the transmission mode selection unit is used to select the transmission mode and output the control signal according to the real-time requirements of the video stream, network bandwidth and stability. The transmission mode includes UDP, TCP and WebRTC; The transmission unit is used to receive the transmission data packet and send the transmission data packet to the video decoding module through the network according to the control signal of the

transmission mode selection unit; the control unit is used to output the control signal according to the network status and adjust the generation and transmission of the data packet. parameter. Further, the decoding module includes: a decoding unit, used to receive the video stream output by the video transmission module, decode the received video stream according to its own codec protocol, and restore the video frame sequence; an information reading unit, used in decoding When a video frame sequence is generated, the node information in it is read and parsed; the information insertion unit is used to receive the decoded video frame sequence and insert decoding node information into the video frame sequence; the watermark embedding unit is used to decode and restore the video. A low-visibility video frame is selected in the frame and a watermark containing each node information is embedded; the output unit is used to output a video frame sequence embedding decoding node information and watermark. Further, the watermark embedding unit includes: a watermark generation subunit, used to generate a watermark containing a check code based on node information; a video analysis subunit, used to analyze the spatial domain information and time domain information of the video frame, and obtain the information of each video frame. importance; the embedding control subunit is used to select video frames whose importance is lower than the threshold according to the importance of each obtained video frame as the target frame for watermark embedding; the watermark embedding subunit is used to embed the generated in the target frame Watermark; among them, video frames whose importance is lower than the threshold are video frames whose image complexity is lower than the threshold. Further, the information insertion unit includes: a coding information supplementing subunit, which is used to supplement the complete coding and decoding node information according to the decoding time and device number; an information steganography subunit, which is used to use steganography technology to add the supplemented node information Extracts are inserted into video frames. Further, the steganography technology is digital watermarking technology. Further, the display module includes: a video display unit, used to receive the decoded video frame sequence, render and display it on the display device; a metadata parsing unit, used to detect and parse the information frames in the video frame sequence, and obtain the It contains the time, location and device number information of each encoding and decoding node; the watermark analysis unit is used to detect the watermark frame in the video frame sequence and analyze the node information watermark; the traceability analysis unit is used to combine the metadata analysis unit and The node information obtained by the watermark analysis unit is fused to form a complete video transmission trajectory from the source to the destination node; the storage unit is used to store the video transmission trajectory information for source traceability query. Further, the traceability analysis unit includes: an information association subunit, which is used to associate the parsed node information based on the time stamp and device number information; a routing generation subunit, which is used to generate a complete route based on the associated node information. Video streaming routing. 3. Beneficial effects Compared with the existing technology, the advantages of the present invention are: (1) By inserting traceable metadata, watermarks and node information into

each link of video processing (collection, encoding, transmission, decoding, display), and collecting and analyzing this information at the display end, the entire processing of video content is realized. The traceability of the process achieves a fine-grained traceability effect;

(2) Through information steganography technology such as digital watermarking, the amount of traceability information contained in the video is increased, and at the same time, encryption and other means are used to improve the security of the information. The comprehensive use of multiple steganography technologies enhances the fault tolerance of traceability information;

(3) By parsing the information inserted into each node of the video processing, and performing information association and route generation, a complete transmission link of the video content from the source to the terminal can be constructed, which enables traceability of the entire video transmission process and helps to track the content spread trajectory and determine the responsible party. In summary, this solution achieves a refined traceability design for the video processing process, which can effectively improve the efficiency of video content traceability, enhance the integrity of traceability results, improve video traceability efficiency, and thereby improve security and confidentiality.

Description of the drawings This specification is further described by way of example embodiments, which are described in detail by means of the accompanying drawings. These embodiments are not limiting. In these embodiments, the same numbers represent the same structures, where: Figure 1 is an exemplary module diagram of a secure traceable video encoding and decoding system according to some embodiments of this specification; Figure 2 is a schematic technical roadmap of a secure traceable video encoding and decoding system according to some embodiments of this specification; Figure 3 is an exemplary flow chart of a traceability analysis process according to some embodiments of this specification; Figure 4 is an exemplary flowchart of a watermark embedding process according to some embodiments of this specification.

Detailed ways **Glossary** The source video stream refers to the original unencoded and compressed digital video sequence input by the system. It is the first link in the traceable video processing process; the source video contains continuous video frames and audio data, stored in a certain encoding format (such as YUV), and also contains video collection parameter information; the source video stream serves as the entire system. The quality of input directly affects the effect of subsequent video processing. The clearer and smoother the source video stream, the more efficient the subsequent compression and encoding. The system generates an encoded video stream that can be transmitted and distributed by encoding, compressing, encrypting, and encapsulating the source video stream. Traceable information such as time also needs to be inserted into the source video. Metadata such as stamp, location, device number, and digital watermark provide basic data for the traceability of transmitted videos; therefore, the source video stream is the source of information for the traceability system, and its collection quality determines the effect of the entire traceability analysis. To sum up, the source video stream is the original video input of the secure and traceable video encoding and decoding system, and its quality directly affects the processing and traceability effect of the entire system. It contains basic video frame information, as well as basic data required for traceability.

Watermark information refers to the information embedded into the source video frame using digital watermark technology during video collection and encoding. This information is hidden in the video signal in an invisible form; the watermark information can include the video collection time, location, device number, user account and other traceable related data; during the video transmission and decoding process, the watermark information is difficult to destroy or delete, can withstand video processing and compression; when tracing the source of transmitted videos, watermark data can be extracted through watermark information analysis technology to assist in confirming the collection source and transmission process of the video; watermark information is different from video metadata, and the metadata is in plain text Direct embedding, while watermark information is embedded in a hidden form, making it more difficult to be illegally modified; watermark technology improves video traceability and anti-counterfeiting. Watermark information and metadata are used together for traceability analysis. In summary, in traceable video systems, watermark information is embedded in videos using digital watermark technology, which serves as an effective supplement to metadata and enhances the reliability and resistance to modification of video traceability. Metadata is auxiliary data attached to the video stream, including description, annotation and supplementary information of the video content; metadata can include video collection time, location, device information, event information, etc., embedded in text format or structured format In the video stream; metadata is generated during video collection and can be inserted or updated at each stage of video processing, or generated by analyzing video content; metadata exists in the video encoding stream and does not affect video decoding and playback, but can be parsed Program reading; when tracing the source of video transmission, metadata can be extracted and parsed from the video stream to obtain information such as the time and place of video transmission; metadata is directly embedded in plain text and is easy to be illegally modified, so it needs to be combined with digital watermark information Combined to improve reliability; metadata makes the video encoding stream self-descriptive and scalable, enriching video information. In summary, metadata provides auxiliary description for traceable video systems and is one of the important information sources for video traceability. Together with watermark information, it improves the effect of traceability analysis. The encoding node contains a digital record of the time, location, operator, equipment and other information of the video encoding process; the encoding node is generated during the video encoding process and inserted into the encoded video stream in the form of metadata or digital watermark; a complete encoding system may include Multiple encoding nodes, such as encoder identification code, encoding standard, encoding time, encoding location, etc.; the decoding process will also generate decoding nodes at the same time, identifying the decoding time and location information; when performing traceability analysis of video transmission, you can start from the encoded video Analysis and extraction of encoding node information from the stream; by associating and merging encoding nodes and decoding nodes, the traceability of the complete video transmission trajectory from encoding to decoding can be completed; encoding nodes are the basic information source for video traceability, and their accuracy and reliability Determines

the effect of traceability. In summary, the encoding node records the key information of the video encoding process, and together with the decoding node constitutes the traceable basic data of the video transmission link. It is one of the core information for realizing video traceability. Compression coding uses a video compression algorithm to encode the source video, remove redundant information, and reduce the amount of video data; commonly used video compression coding standards include H.264, H.265, etc., which can significantly reduce the amount of video data; compression coding can Adjust the bit rate and quality parameters of the generated video according to network bandwidth and other conditions; the compressed and encoded video can be efficiently transmitted and stored, but compression noise is also introduced; in a traceable system, the compression and encoding process needs to retain necessary metadata and watermark information for subsequent traceability; the decoding link restores the compressed and encoded video according to the corresponding encoding and decoding algorithm; using appropriate compression encoding standards and parameter control, high compression rates and good visual quality can be obtained. Encapsulation is the process of packaging compressed and encoded video streams and audio streams into container formats; commonly used video encapsulation formats include MP4, MKV, AVI, TS, etc. These formats define the storage and packaging methods of video, audio and metadata; the encapsulation format also defines the packaging method of the code stream. For example, MP4 uses MP4 file format and MPEG-4 encoding; encapsulation adds synchronization information and timing to the video encoding stream. information, fault-tolerant information, etc., which facilitates storage and transmission; in the traceability system, the metadata and watermark information in the video need to be retained during the encapsulation process to package the encapsulation container; the decapsulation process parses the container and extracts the video stream and audio stream and metadata to provide data for subsequent decoding; encapsulation shields the complexity of code stream transmission, making storage and transmission easier. In summary, video encapsulation realizes the packaging and formatting of compressed and encoded bit streams, and is a key link in the video processing process. The propagation link includes the time and location information of all processing links such as video collection, encoding, encryption, encapsulation, transmission, and decoding; the processing of each link will generate metadata or watermark information nodes, and these node information are connected in series to form a complete propagation link; The propagation link identifies the movement trajectory of the video and can track the source and transmission flow of the video. When tracing the source of the video content, it is necessary to analyze the node information in the video and reconstruct the complete propagation link; the propagation link provides the video Technical guarantee of authenticity, traceability and source verifiability; if the information in any part of the transmission link is destroyed, it will lead to the failure of traceability analysis; a complete and accurate transmission link is the basis for video traceability and is also the basis for judging the video important basis for authenticity. In summary, the transmission link records the complete flow path of video content from source to terminal by encoding the time and location nodes of each processing link, which is the key

information to achieve video traceability. The video output device is located after the system decoding module and is used to render the decoded digital video signal and convert it into visible picture output; common video output devices include LCD monitors, projectors, TVs, etc. They accept different forms of video input signals; the video output device includes a video interface, a video decoder, a display panel/projection component, etc., to convert digital video into a visible picture; in a traceability system, the output device displays a video containing metadata and watermarks. The complete video content may also include traceability query information; the performance of the output device determines the quality of the video picture, such as resolution, color reproduction, dynamic range, etc.; different output devices are suitable for different usage scenarios, such as projectors for conferences, TVs for home entertainment; the output device is the last link in the video processing chain, facing users directly, and its performance is directly related to the user's perception. In summary, the video output device is the terminal picture output link of the traceability system. It converts digital video into visible pictures through the analysis and rendering of the video stream, which directly affects the user experience. Coding standards stipulate the specific algorithm flow, syntax format, implementation methods, etc. of video compression coding; common video coding standards include H.264, H.265, AV1, etc. New standards usually improve encoding efficiency while reducing bit rates; encoding standards ensure the standardization of the syntax and process of video compression, making encoded videos standard-compatible; different encoding standards are optimized for specific application scenarios, such as high-definition video, video Surveillance, network video, etc.; in a traceable system, choosing the appropriate encoding standard can balance video quality and encoding efficiency; the decoding end uses the same standard as the encoding end to correctly parse and restore compressed video; with the improvement of hardware performance, A new generation of encoding standards is activated, such as the current H.265. H.264 is an international standard for digital video compression and coding jointly issued by ITU and ISO. Also known as MPEG-4 Part 10; compared with old standards such as MPEG-2, H.264 has a compression rate that is approximately 2 times higher at the same image quality; H.264 achieves higher coding efficiency through more complex compression algorithms. However, the coding complexity also increases; H.264 is widely used in video surveillance, high-definition digital TV, online video, mobile devices and other fields; in traceability systems, H.264 can significantly reduce the bit rate while ensuring video quality; H.264 can also retain necessary metadata and watermark information in the video to achieve traceability; only decoders that support the H.264 standard on the decoding end can correctly parse and render H.264 encoded videos. In summary, H.264 is one of the current mainstream video coding standards. It can achieve high-efficiency video compression and coding in traceability systems while ensuring the integrity of video quality and traceability information. H.265 is the latest video coding standard released by ITU and ISO based on the H.264 standard. Compared with H.264, H.265 has a compression rate that is approximately 30%-50% higher under the same visual quality; H.265 Higher coding efficiency is achieved through more complex compression algorithms, but the coding complexity is also greatly increased; H.265

supports efficient compression of high-resolution video content such as 4K and 8K; in a traceable system, H.265 can ensure quality On the premise of significantly reducing storage and transmission costs; H.265 encoding can also retain metadata and watermark information in the video to achieve traceability; the decoding end needs a decoder that supports H.265 to correctly parse H.265 compressed video ;Currently H.265 has not yet been fully popularized, and the cost of hardware adoption is high. Denoising is the use of digital signal processing technology to eliminate or reduce the noise component in video signals; video sources are susceptible to interference from various types of noise during collection and transmission, such as electronic noise, coding compression noise, etc.; denoising can effectively reduce the noise content of video sources. Existing random noise and sharp noise can improve image quality; commonly used denoising algorithms include median filtering, wavelet denoising, adaptive denoising, etc. The intensity of denoising needs to be well controlled; in traceability systems, proper denoising can improve the efficiency of video compression and encoding; however, denoising may also cause loss of video details, and it is necessary to ensure that the features required for traceability are not blurred; through hardware optimization and parameters Control can reduce noise while retaining key information of the video. Color correction mainly focuses on color restoration and enhancement to address color distortion problems in video collection and encoding; video sources often suffer from color deviations, color fading and other problems due to lighting and equipment factors. Color correction can correct these defects; commonly used color correction techniques include white balance, color mapping, gamma correction, etc., which can correct the contrast, saturation, color temperature, etc. of the video; proper color correction can make the video color more realistic and vivid, and improve vision. Effect. However, it should not be overly beautified and affect the authenticity; in the traceability system, color correction parameters must be controllable to avoid destroying the original tonal characteristics of the video; color correction also needs to ensure that the relative relationship between the colors on the face and the scene remains unchanged, so as to facilitate Traceability and comparison; by selecting the appropriate color correction algorithm and intensity, the video quality can be improved without affecting the authenticity of the video content. The device number can be a numeric serial number or a complex code containing letters and special characters; the device number uniquely identifies each device in the video processing link, such as cameras, encoders, servers, etc.; through the device number, multiple devices can be distinguished. devices of the same model, and monitor and trace the source of each device; device number information can be inserted into the video stream in the form of metadata, or embedded through invisible watermarks; when tracing the source of video transmission, the device number is used to confirm the source and origin of the video. The key information of the processing process; the device number can also bind the video content with the information of the device user, as a basis for responsibility traceability; through the extraction and analysis of the device number, it helps to quickly lock the generation and processing of video content. In summary, the unique device number identification plays a key role in tracking and locating the source of the video in the traceability system, and is a basic information

component to achieve traceability. The working mode includes the encoding mode, decoding mode, and transmission mode of the video codec; the encoding mode determines the encoding standard, compression rate, encoding speed and other parameters of the video; the decoding mode determines the video analysis method, processing priority, and playback control and other parameters; the transmission mode determines the settings of the video transmission protocol, transmission priority, network QOS and other parameters; by selecting and switching different working modes, the dynamic optimization and control of the video processing link can be achieved; in the traceability system, different working modes need to ensure the integrity of video traceability information; the switching process of working modes also needs to be logged, so that parameter changes can be tracked during traceability analysis; reasonable working mode configuration can optimize the traceability system while ensuring video quality. performance. Node information includes the time, location, equipment, operators and other data of video collection, encoding, storage, transmission, decoding and other processing nodes; node information is embedded in the video stream in the form of metadata or digital watermarks; every link of the video processing process Corresponding node information will be generated. All node information chains are connected in series to form a complete video processing link; when tracing the source of video content, it is necessary to extract and parse the node information in the video stream; node information can help quickly determine the generation time, location, circulation path and operation of the video History; node information directly determines the degree of sophistication of traceability. The more detailed the node information, the more accurate and reliable the traceability will be; protecting the integrity and security of node information is an important part of the traceability system. In summary, node information is the basic data for video traceability, and its accuracy directly affects the effect of traceability. Video frame sequence, video generates dynamic images by transmitting multiple image frames per second; each image frame contains independent video content information, and continuous image frames constitute a video sequence; the video plays frames at a rate of 25 to 60 frames per second sequence to produce a visual motion effect; video compression coding reduces frame sequence redundancy through inter-frame prediction and other methods; in a traceable system, introduces timestamp or frame coding information to the video frame sequence; by analyzing the details of the video frame sequence Changes can be used to perform traceability analysis of video content; for example, the time and place of the video can be verified through changes in scenes and characters in consecutive frames; protecting the integrity of the video frame sequence is crucial for traceability. In summary, the video frame sequence contains video time information and content detail information, and is the basic data unit to achieve accurate video traceability. The encoder compresses and encodes the collected video source content and outputs a compressed bit stream; common encoders include software encoding libraries, hard-coded cards, embedded encoding chips, etc.; the encoder compression algorithm needs to implement video compression standards, such as H. 264, H.265, etc.; the encoder optimizes the video compression rate while controlling the stream quality and encoding speed; in a traceable system, the encoder also needs to retain

key metadata while compressing; the encoder parameter settings directly affect the video Compression quality and bit rate; the performance of the encoder determines the speed of video encoding and affects the entire video processing link; choosing the appropriate encoder and parameter settings is crucial to the traceability system. In summary, the encoder is responsible for efficient compression of video content in a traceable system, and its performance is directly related to video quality and encoding and decoding speed. Symmetric encryption algorithms use a single key, which is used for data encryption and decryption; common symmetric algorithms include DES, AES, RC4, etc. Among them, AES has higher security; symmetric encryption is fast and the algorithm is relatively simple, which is suitable for encrypting large amounts of video content; encrypting video data can prevent unauthorized access and ensure data security; in a traceability system, it is necessary to encrypt the content contained in the video metadata information; however, encryption will also introduce algorithm noise, and the impact on video quality needs to be controlled; the encryption key needs to be securely managed to prevent the key from being leaked to unauthorized users; choosing a safe symmetric encryption algorithm can ensure speed while ensuring , effectively protecting the security of videos and metadata. Asymmetric encryption uses public key encryption and private key decryption, which enhances the flexibility of key management; commonly used asymmetric algorithms include RSA, ECC, etc. It has high security but is slower than symmetric encryption; asymmetric encryption can be used to encrypt video metadata, as well as digital signature and certificate management; the public key can be made public, and the private key is held by the video content owner, with stronger control capabilities; In a traceable system, asymmetric encryption can be used to protect the authenticity of video processing node information; a device authentication mechanism based on digital certificates can also be established to ensure the security of the traceable system; asymmetric encryption algorithms are complex in calculation and are not suitable for large amounts of video data Encryption can be used in conjunction with symmetric encryption; rational application of asymmetric encryption can strengthen content protection and permission control of the traceability system. The video transmission protocol standard refers to the data packet format and interaction rules used for video transmission in the transmission module. Common video transmission protocol standards include: RTP, Real-Time Transport Protocol, a transmission protocol used to transmit audio and video over the Internet. It can process source identification, sequence number, time stamp, etc. of multimedia data to achieve real-time data transmission; RTSP, real-time streaming protocol, is an application layer protocol used to control the start, pause and other operations of real-time data transmission. Often used in conjunction with RTP protocol; HTTP, Hypertext Transfer Protocol, application layer protocol, used for Web video transmission. Supports various encryption mechanisms to improve security; HLS-HTTP real-time streaming protocol, proposed by Apple and designed based on HTTP protocol, is used for streaming media transmission; MPEG-DASH, dynamic adaptive streaming protocol, is used for the transmission of high- definition multimedia content , supports dynamic code rate switching; WebRTC, web real-time communication technology, supports browser-to-browser video

streaming; SRT, a safe and reliable transmission protocol, is used for low-latency real-time video streaming. In this secure and traceable video encoding and decoding system, appropriate video transmission protocol standards can be selected and used based on actual needs and network conditions to achieve efficient and reliable video stream transmission, thereby ensuring the smooth progress of video traceability analysis. The code stream is a digital video bit stream sequence compressed by video coding and contains encoded video frame information. In this safe and traceable video encoding and decoding system, the main code streams include: source video code stream, the digital video bit stream output by the video acquisition module that meets the coding standard requirements; encoded code stream, the video encoding module after compression and encoding Digital video bit stream output; transmission code stream, the digital video bit stream used for transmission after the transmission module encapsulates and encrypts the encoded code stream; decoding code stream, the decoding module receives and decodes the code stream output by the transmission module to obtain the digital Video bit stream; output stream, digital video bit stream containing watermark information output and displayed by the display module. By inserting metadata and node information at different links and embedding watermarks in the output code stream, the propagation link of the code stream can be tracked and the source of the video content can be traced. As the input and output of the system, the code stream carries important information required for video traceability. Generally speaking, the code stream is a compressed and digitalized video data stream, which plays a key role in video storage, transmission and display in this secure and traceable video encoding and decoding system. Obfuscation processing refers to the confidentiality processing performed by the encryption unit in the transmission module on the encoded video frame sequence. Its typical purposes include: protecting the security of video content and preventing unauthorized third parties from directly obtaining and using the original video; avoiding information leakage that may occur when directly transmitting encoded video frame sequences; improving the security of subsequent decoding so that only authorized The receiver can decode correctly. Common obfuscation processing methods include: using symmetric or asymmetric encryption algorithms to encrypt full frames or selected frames of video frames; disrupting the visual content in the video frames and destroying the original visual information; inserting interference signals or pseudo-images into the video frames. Random noise sequence; scramble the video frame number or time information; use network security technology to encapsulate and hide the transmission traffic. Through obfuscation, the security of video transmission and decoding can be improved and information leakage can be avoided. It is one of the necessary security measures in the safe and traceable video encoding and decoding system. Only the decoder can correctly parse the obfuscated video frame content. The data packet is the transmission unit after the transmission module segments and encapsulates the encoded video stream. According to different video transmission protocol standards, data packets can contain the following information: video data payload, including compressed and encoded video frame information; packet header, including packet information, such as packet sequence number, timestamp, data length, etc.; source and

destination addresses , identifies the sender and receiver addresses of the data packet; verification information, to ensure the correctness of the data packet transmission; control information, used to regulate and control the transmission of the video stream; encryption information, to encrypt the video data load to ensure transmission security . Data packets are data transmission units that are optimized and encapsulated by the transmission module based on network conditions, reliability requirements, transmission protocol standards, etc. By controlling the content and transmission of data packets, safe and reliable transmission of video streams can be achieved, providing guarantee for subsequent analysis and traceability analysis of video frames. The generation and transmission of data packets are critical to network performance of secure and traceable video codec systems. Network bandwidth refers to the maximum data throughput or communication capacity of the network connection during video transmission. When selecting the video transmission mode, the transmission module of the system needs to consider the network bandwidth, which mainly includes: the network connection type (wired/wireless) determines the physical basis of the network bandwidth; the network bandwidth directly affects the selection of code rate control parameters during video encoding; High-bandwidth networks can transmit high-quality, low-latency video streams; low-bandwidth networks need to reduce the bit rate and packet size to ensure basic video transmission quality; dynamically monitor network bandwidth changes to achieve adaptive adjustment of bit rates; different transmission Modes (UDP/TCP/WebRTC) have different requirements for network bandwidth. Sufficient and dynamically controllable network bandwidth is the basis for ensuring the system's reliable transmission of video streams and smooth video analysis and traceability. Establishing reasonable network connections, evaluating and controlling network bandwidth are important aspects of a secure and traceable video codec system. Network stability refers to the reliability and anti-interference ability of the network connection during video transmission. When selecting the transmission mode of the system's transmission module, network stability needs to be considered. The main issues include: whether the network connection is prone to breakage and jitter, which directly affects the smoothness of video transmission; compared with wired networks, wireless networks are more susceptible to environmental interference and stability. Poor; network congestion will cause video packet loss, and congestion control is required; choosing TCP transmission can obtain a positive response and retransmission mechanism, improving reliability; although UDP transmission has lower real- time requirements, network stability directly affects packet loss rate; you can use network quality detection to dynamically switch transmission modes or adjust encoding parameters; deploy SDN networks to optimize and control data flows and improve stability. Good network stability is the basis for the system to meet user experience requirements, and is also an important factor to ensure smooth video analysis and traceability. It is necessary to evaluate and improve network stability. The transmission mode refers to the data packet transmission method selected by the video transmission module based on the characteristics of the video stream and network conditions. The main transmission modes mentioned in the system include: UDP mode, a transmission mode based on the User Datagram Protocol, which has the advantages of

strong real-time performance and simple protocol, but poor reliability; TCP mode, a transmission mode based on the Transmission Control Protocol, has Reliable transmission, flow control and other features, but the transmission delay may be high; WebRTC mode, a browser-to-browser real-time multimedia transmission technology, based on UDP, can achieve low-latency transmission. The transmission module can select the appropriate transmission mode based on video real-time requirements, network bandwidth and stability, etc., to obtain a better user experience. At the same time, the system can also dynamically switch transmission modes to adapt to changes in network conditions. Reasonable transmission mode selection is an important means to ensure system transmission efficiency.

UDP (User Datagram Protocol) is a connectionless transport layer protocol. When used as a transmission mode in this system, the main features of the UDP protocol include: providing connectionless and unreliable services, transmitting data packets without confirmation and retransmission; low transmission delay, and more suitable for applications with high real-time requirements. Video stream transmission; the protocol is simple and has high transmission efficiency, small header overhead, and supports one-to-many network multicast; datagrams may be lost or out of sequence, and reliable transmission is not guaranteed; the application layer needs to implement mechanisms such as flow control and retransmission to improve reliability; lower security, vulnerable to denial of service type attacks; transmission rate depends on network bandwidth and stability. In the system, UDP mode transmission can be appropriately used based on the video stream characteristics and network conditions to obtain the benefits of low latency and high efficiency, but reliability issues need to be weighed.

TCP (Transmission Control Protocol) is a connection-oriented and reliable transport layer communication protocol. As a transmission mode in this system, the main features of the TCP protocol include: establishing a connection through a three-way handshake, transmitting data point-to-point, and providing reliable delivery services; supporting flow control and congestion control to avoid network congestion; using a confirmation response mechanism to support Timeout retransmission ensures the reliability of data transmission; maintains the transmission sequence and handles data packet loss; is connection-oriented, has high security and is difficult to be attacked; the transmission delay may be high, and is not as suitable as UDP for applications with extremely high real-time requirements scenario; the header overhead is large and the transmission efficiency may be lower than UDP. The system can use the reliable transmission characteristics of the TCP protocol to ensure the integrity of the video stream according to requirements, but the issue of transmission delay needs to be weighed.

WebRTC (Web Real-Time Communication) is a real-time communication technology that supports browser-to-browser real-time voice, video and data communication. As a transmission mode in this system, the main features of WebRTC include: realizing low-latency real-time multimedia transmission based on UDP protocol; supporting video encoding such as H.264, audio encoding such as Opus, etc.; transmitting data through P2P without the need for a server Relay; directly implement media negotiation, encoding and decoding functions within the browser; penetrate firewalls and NAT through STUN, TURN and other protocols; widely used, can realize point-to-point real-time video

calls in Web pages; high security, supports DTLS encryption transmission. WebRTC can be used as the transmission mode in the system to transmit real-time video streams in P2P form, achieving low latency and high efficiency, and ensuring a certain level of security. Transmission parameters refer to some settings that the system can control and adjust when transmitting video streams to achieve better transmission effects. Common transmission parameters include: transmission protocol, selection of UDP/TCP/WebRTC and other protocols; encryption algorithm, the type of algorithm used to encrypt the video stream; transmission rate, the amount of data transmitted per unit time; packet size, each data packet contains PAYLOAD data size; buffer size, the queue or buffer size used by the receiving end to cache data packets; packet loss processing, retransmission or compensation strategy when data packets are lost; delay control, end-to-end transmission delay time control; errors Verification, verifying data to check for transmission errors; flow control mechanism, flow control algorithm, to avoid network congestion. Adjusting these transmission parameters can optimize the encoding performance, transmission stability, delay characteristics, etc. of the video, improve the transmission quality of the system, and safeguard subsequent video analysis and traceability. The codec protocol is the standard or specification that the video codec module follows when compressing, encoding and decoding the video stream. The main codec protocols in this system may include: H.264/AVC, a widely used video compression coding standard with high coding efficiency and support for multiple resolutions; H.265/HEVC, a new generation of high-efficiency video coding standard, Compared with H.264, it has more than 50% compression efficiency improvement; VP8, an open source video encoding format proposed by Google; VP9-upgraded version of VP8, using more encoding tools to improve compression rate; AV1, launched by AOMedia Alliance Open source video coding format; MPEG-2, the video coding standard used in early digital TV broadcasts; MPEG-4, a coding standard that provides better compression performance and supports object encoding. Choosing an appropriate codec protocol can effectively improve the video compression rate, reduce storage and transmission overhead, and thereby improve system performance. But there is also a need to balance the relationship between encoding complexity and video quality. Low-quality video frames refer to video frames with poor quality reconstructed after compression through video coding. Its characteristics usually include: the allocated code rate or number of bits is small, resulting in serious image distortion; the frame content is complex, and the encoder selectively reduces the quality of the frame to maintain the image quality of adjacent frames; there is a large area of blurry area or mosaic Effect; coding distortion is obvious in video areas with severe motion changes; accumulation of transmission errors in intra-frame prediction coding algorithms causes quality degradation; video quality loss caused by packet loss and frame loss in network transmission; image damage at the decoder end caused by limited computing resources. Check code is a redundant code used to verify and verify the correctness of data transmission or storage. The main purpose of adding check codes in the video stream transmission and storage process is to detect and correct data errors during the transmission process and improve reliability; to detect data damage on the storage medium through verification; to verify the integrity and

consistency of the data, Protect against unauthorized tampering. Common check code algorithms include: parity check, which is simple and efficient and can detect single errors; cyclic redundancy check (CRC), which can detect sudden errors and is widely used; message authentication code (MAC), which prevents message forgery ; Secure Hash Algorithm (SHA), which can detect any modification; Checksum, which is a simple sum and remainder. Spatial domain information refers to the visual information contained in the spatial domain of the video signal. Spatial domain information will be utilized and processed during the video encoding and compression process, mainly including: motion prediction will determine and utilize the movement information of pixel blocks in the spatial domain; transformation coding will convert the spatial domain pixels into the frequency domain for encoding; quantization will retain the spatial domain visual information The main components; video filtering operations directly process pixels in the spatial domain; video enhancement can also be adjusted in spatial domain contrast, chroma and other parameters; video watermarks, etc. are directly embedded in spatial domain pixel information; video compression standards will utilize spatial domain redundancy. Decorrelation operation; decoding and reconstructing frames also reconstructs video content in the spatial domain. Time domain information refers to the information contained in the video signal changing with time. Time domain information is utilized and processed during the video encoding and compression process, which mainly includes: calculating motion vectors between consecutive frames and judging changes in time domain motion; using temporal redundancy to remove static backgrounds; adjusting the GOP structure to control key frame insertion time ;Adjust the frame rate and code rate control parameters of the encoder; obtain scalability through temporal level prediction coding; adjust the code stream in real time according to the time domain information during transmission; time domain filtering can smooth the video time domain noise; time-process the video clips Domain cropping and splicing; judging video content through time domain statistical analysis. Image complexity usually refers to an indicator that describes the spatial information and temporal information of a video image, and can be used to evaluate the complexity of a video image. Video frames with low image complexity are used as target frames for watermark embedding, which can avoid embedding watermarks in high-complexity image areas, thereby reducing the impact of watermarks on video quality. The calculation of image complexity can consider the following aspects: Spatial complexity: reflects static information such as spatial details and edges of the image, which can be represented by intra-block standard deviation, edge density, etc.; Time complexity: reflects dynamic change information of the image sequence , can be represented by motion vector size, residual information, etc.; Texture complexity: reflects the directional information of the image, which can be represented by gradient histograms, etc.; Object complexity: reflects semantic information such as the number and type of subject targets in the image; Noise Complexity: reflects the noise component of the image. Steganography technology can hide and embed traceability information such as encoding/decoding nodes into video frames, which does not affect the display of the original video, but can be extracted through analysis. Common steganography techniques include: least significant bit (LSB) embedding:

embedding information bits in the least significant bits of video pixels; spectrum expansion, hiding information in high-frequency components of the video signal; quantization index modification, adjusting video compression coding Quantization parameters, embedding information during the quantization process; motion vector modification, adjusting the motion vector information of video compression encoding for steganography; jitter blur, embedding information by introducing video jitter; statistical model embedding: utilizing the statistical characteristics of video compression for steganography ; Deep learning steganography, using deep neural networks for steganography. Digital watermarking is a technology that embeds information covertly into signals. It can insert auxiliary information into video signals for subsequent detection and extraction without affecting the main body of the signal. In this traceability system, digital watermark technology can be used to embed encoding, decoding and other node information into video frames in a steganographic manner for traceability tracking. The main characteristics of digital watermarks include concealment, since the watermark will not affect the visual effect after being embedded; robustness, the watermark can resist common signal processing operations; security, the watermark has a certain anti-tampering ability; capacity, the watermark can be embedded in a certain number of Valid information. The implementation methods of digital watermarking include: spatial domain method, which is directly embedded in pixel samples; frequency domain method, which is embedded in the transformation domain, such as DCT, DWT, etc.; compression domain method, which is embedded in the video compression encoding process; geometric attack recovery method, which improves Resistance to geometric attacks; fusion embedding method, combined with multi-domain information for embedding. Rendering refers to the process of converting a sequence of decoded video frames into a video signal that can be played on a display device. Specifically, the rendering process includes the following steps: color space conversion, which converts the YUV or RGB color space of the video frame to the color space used by the display device, such as sRGB; scan conversion, which converts the progressive scanning order of the video to the display device's Progressive or interlaced format; scaling, scaling the resolution of the video to the native resolution of the display device; de-interlacing, de-interlacing the interlaced video signal; subtitle overlay, superimposing subtitle images or text onto the video; Connector conversion converts digital video signals into analog signal interfaces used by display devices; display drivers send out correct synchronization signals to drive the display panel to display video images. Through the rendering process, the decoded and recovered video sequence can be played correctly on the display device, providing users with a good viewing experience. A timestamp is an identifier that digitally encodes the time an event occurred. In this system, when each processing node operates on a video frame, it will insert encoding timestamp or decoding timestamp information to identify the processing time of the video data. These timestamp information are used for subsequent traceability analysis to track the video's propagation link. Transmission routing refers to the path that data packets take when transmitted from the source address to the destination address in the computer network. The system's traceability analysis unit can generate a complete transmission route through which the data packets of

the video stream pass based on the analyzed node information, reflecting the propagation process of the video stream. Example In order to explain the technical solutions of the embodiments of this specification more clearly, the accompanying drawings needed to be used in the description of the embodiments will be briefly introduced below. Obviously, the accompanying drawings in the following description are only some examples or embodiments of this specification. For those of ordinary skill in the art, without exerting creative efforts, this specification can also be applied to other applications based on these drawings. Other similar scenarios. Unless obvious from the locale or otherwise stated, the same reference numbers in the figures represent the same structure or operation. It should be understood that "system", "device", "unit" and/or "module" used in this specification is a method used to distinguish different components, elements, parts, parts or assemblies at different levels. However, said words may be replaced by other expressions if they serve the same purpose. As shown in this specification and claims, unless the context clearly indicates an exception, words such as "a", "an", "an" and/or "the" do not specifically refer to the singular and may also include the plural. Generally speaking, the terms "comprising" and "comprising" only imply the inclusion of clearly identified steps and elements, and these steps and elements do not constitute an exclusive list. The method or apparatus may also include other steps or elements. Flowcharts are used in this specification to illustrate operations performed by systems according to embodiments of this specification. It should be understood that preceding or following operations are not necessarily performed in exact order. Instead, the steps can be processed in reverse order or simultaneously. At the same time, you can add other operations to these processes, or remove a step or steps from these processes. Figure 1 is an exemplary module diagram of a secure traceable video coding and decoding system according to some embodiments of this specification. As shown in Figure 1, a secure and traceable video coding and decoding system 100 includes: a video collection module 110; Video encoding module 120, transmission module 130, decoding module 140 and display module. 150 The video collection module 110 is used to collect the source video stream and insert watermark information and metadata into the source video stream. The video encoding module 120 is configured to receive the source video stream, insert encoding node information into the source video stream, and perform compression encoding on the video stream after inserting the encoding node information. The transmission module 130 is used to encapsulate and transmit the encoded video stream output by the video encoding module 120. The decoding module 140 is configured to receive the video stream output by the transmission module 130, decode the received video stream, and insert decoding node information and watermark into the decoded video stream. The display module is used to display the video stream output by the decoding module 140, parse node information and watermarks therefrom, and generate a transmission trajectory of the video stream. Among them, the display module includes a traceability analysis unit, which is used to obtain the decoded video stream, analyze the node information and watermarks in the video stream, and track the image propagation link. Through this secure and traceable video encoding and decoding system, necessary information can be inserted into each node of video collection,

encoding, transmission, and decoding, and the information can be parsed at the display end to trace the transmission trajectory of the video stream and ensure the authenticity of the video content. and traceability. Among them, the video collection module 110 inserts watermarks and metadata into the source video stream. The watermark contains information such as time, location, device number, etc., and the metadata contains the results of detection and identification. This provides key information for subsequent source tracing; the video encoding module 120 continues to insert node information such as encoding time, location, and device number into the source video stream, and performs compression encoding. Compression encoding reduces the amount of video data and facilitates transmission. The inserted node information is also the key to source tracing; the transmission module 130 encapsulates and transmits the video stream. Encapsulation improves transmission efficiency, and the transmission sends the video to the decoding end; the decoding module 140 decodes the received video stream and restores the video frame. At the same time, node information such as decoding time, location, and device number is also inserted into the video stream; in the decoded video stream, the watermark embedding module selects an appropriate frame to embed a watermark containing each node information. The concealment of watermarks makes the information less susceptible to damage; the display module generates a complete video transmission trajectory by analyzing the node information and watermarks in the video stream. Different forms of information are cross-verified to improve traceability accuracy. Specifically, the video collection module 110 is in the first stage of video stream generation in the secure and traceable video encoding and decoding system. Its main function is to collect, enhance and feature identify the source video stream to prepare for subsequent encoding, transmission and traceability analysis.

Generate optimized source video streams. Component units: The video collection module 110 includes a video source unit, a coding standard selection unit, a video collection unit, an image enhancement unit, a watermark information unit, a metadata generation unit and a collection control unit. Video source unit: used to obtain the original video signal generated by the camera or computer desktop. Coding standard selection unit: used to select H.264, H.265 and other video coding standards based on the format and system requirements of the source video signal. Video acquisition unit: used to collect source video signals according to the selected encoding standard and output source video streams that meet the standard requirements. Image enhancement unit: used to perform image quality enhancement processing such as denoising and color correction on the source video stream. Watermark information unit: used to insert a digital watermark containing video collection time, location, device number and other information at a preset position in the source video stream. Metadata generation unit: used to regularly extract key frames from the source video stream and generate metadata containing information such as target detection and scene recognition through analysis. Collection control unit: used to control the working mode of the video collection module 110 according to the network status and user instructions. Workflow: Obtain the original video signal; select the encoding standard according to the source signal; collect the source video stream according to the standard; perform image enhancement on

the source stream; insert watermark information into the source stream; generate metadata describing the content of the source stream; control the working mode according to the network and instructions. In the source video stream generation stage, the video collection module 110 realizes feature identification of the source video stream by inserting watermarks, metadata and other information, and provides basic guarantee for subsequent video encoding, transmission and traceability analysis. More specifically, when the watermark information unit inserts a digital watermark into the source video stream, its preset position can be set in different ways: fixed position: such as the upper left corner or lower right corner of the video sequence. This method is simple to operate, but easy to be destroyed; random position: insert watermarks at random positions in the video frame every certain number of frames. This method improves robustness, but the detection is complex; adaptive position: automatically selects blurry areas or noise areas to insert watermarks based on video content, which can reduce visual impact. But content analysis is needed. The source video frame undergoes frequency domain conversion, and according to the visual complexity of the block, blocks with lower complexity are adaptively selected to embed watermarks, while random offsets are introduced to increase robustness. On the decoding end, detection is performed based on the key and random seeds, which can achieve low visual impact and robust watermark detection; the watermark payload information can carry the collection timestamp, location information, device number, etc., and add a check code to verify the watermark. effectiveness. More specifically, the metadata generation unit needs to regularly extract key frames in the source video stream to analyze and generate metadata. The "regular" here can have several specific technical settings: fixed interval: for example, extract a key frame every 10 seconds or every 50 frames. This method is simple to implement, but cannot adapt to changes in video content; adaptive interval: dynamically adjust the extraction interval of key frames according to changes in video content, such as increasing the extraction frame rate when a scene switch is detected. This method can ensure that key frames capture changes in video content, but the implementation is more complex; event-driven: key frame extraction is triggered when specific events in the video stream (such as objects entering the scene, obvious motion, etc.) are detected. This method can extract important content changes. More specifically, a key frame is a reference frame in a video sequence that represents the entire picture. It is independent of other frames and does not depend on the information of other frames. Video compression coding usually uses key frames as the basis to encode ordinary frames in between. The settings of key frames need to be considered: indicating the integrity of the picture content: key frames need to fully reflect the current picture content and provide a complete reference for subsequent ordinary frame encoding; encoding compression efficiency: direct encoding of key frames will occupy a large bit rate, setting Too many keyframes will reduce compression efficiency; convenience of random access: keyframe settings must allow decoding access from any location; processing complexity: the number of keyframes directly affects the amount of encoding calculations; typical keyframe setting strategies are: Timing insertion: For example, a key frame is inserted every 5 seconds, every 50 frames, etc.; Scene switching detection: When a

significant change in the video scene is detected, a key frame is inserted; Motion detection: When the image content changes in a large area, the determination For scene changes, perform keyframe insertion. More specifically, the video source unit provides original video signal input and is the input source of the entire module; the encoding standard selection unit selects the encoding standard according to the characteristics of the source signal so that subsequent processing meets the standard requirements; the video acquisition unit bases the selected encoding standard on The source video signal is digitally collected and sampled to output a digital video stream; the image enhancement unit performs signal denoising and color correction on the digital video stream to improve video quality; the watermark information unit embeds watermarks in the video stream to realize digital identification of video content. Watermark technology ensures that the watermark is not easily removed and enhances the traceability of the video; the metadata generation unit analyzes the video content and extracts semantic information to generate metadata, which is used to describe the video scene and target, and enhances the video semantic identification with the watermark; the collection control unit External conditions such as the network dynamically adjust module working parameters to ensure the reliability of video collection; each unit works together to ensure video quality while using technical means such as watermarks and metadata to trace the source of the video stream, laying the foundation for subsequent processing. . Metadata and watermark technology enable digital feature extraction of video content. In summary, the video collection module 110 collects feature identifiers from the source video, and realizes reliable and traceable video stream generation through the organic cooperation of each unit. Specifically, the video encoding module 120 performs coding, compression, encryption, and encapsulation processing on the video stream based on the source video stream output by the video collection module 110 to generate a transmissible encoded video stream. Component unit The video encoding module 120 includes an information insertion unit, an encoding unit, an encryption unit and an encapsulation unit. Information insertion unit: used to insert encoding node information including encoding time, location, device number and other information into the source video stream. Encoding unit: Used to select the H.264/H.265 standard to encode the video stream according to the network status. The code rate control and variable block size parameters can be adjusted to achieve encoding compression. Encryption unit: used to encrypt the encoded video stream using symmetric/asymmetric encryption algorithms. Encapsulation unit: used to encapsulate the encrypted video stream according to the requirements of the video transmission protocol. Work flow: Insert encoding node information into the source video stream; select encoding standards and parameters according to the network status; encode and compress the video stream; use encryption algorithms to encrypt the encoded stream; complete stream encapsulation according to the transmission protocol requirements. The video encoding module 120 is in Based on the source video stream, through coding compression, encryption and encapsulation processing, the security and transmission efficiency of the video stream are improved, and coding-optimized streaming media data is provided for subsequent video transmission. More specifically, the information insertion unit

embeds encoding node information in the source video stream, which provides node identification for subsequent traceability; the encoding unit selects appropriate encoding standards and control parameters according to the network status, and uses the encoding algorithm to perform lossless or lossy processing on the video data. Compression, reducing the video bit rate and improving transmission efficiency; the encryption unit applies encryption algorithm technology to encrypt the encoded video stream to ensure the confidentiality and security of the video content. The encryption algorithm controls the encryption strength; the encapsulation unit encapsulates the format of the video stream according to the transmission protocol requirements and generates streaming data packets that can be transmitted on the network; the encoding unit compresses the video, reduces the bit rate, and improves transmission efficiency. The encryption unit keeps the content secure. The packaging unit implements transmission adaptation. The three cooperate to optimize video encoding to adapt to network transmission; the information insertion unit cooperates with the encryption and encapsulation unit to ensure efficiency and security while also retaining the traceable information of the video; each unit functions according to its own technical principles and coordinates with each other, jointly complete the optimization processing of video encoding, encryption, and encapsulation, so that the video stream can be transmitted over the network efficiently and safely. In summary, the video encoding module 120 realizes encoding optimization of the source video stream through the collaborative application of each unit technology, and generates an encoded video stream that can be transmitted confidentially. Specifically, the transmission module 130 encapsulates and transmits the encoded video stream output by the encoding module to implement network distribution of the video stream. Component units: The transmission module 130 includes an encryption unit, an encapsulation unit, a transmission mode selection unit, a transmission unit and a control unit. Encryption unit: used for additional encryption and obfuscation of the encoded video stream. Encapsulation unit: used to encapsulate the encrypted video stream into data packets for network transmission. Transmission mode selection unit: used to select transmission modes such as UDP, TCP or WebRTC based on the real-time requirements of the video stream, network bandwidth and other conditions. Transmission unit: used to send the encapsulated video data packets to the decoding end through the network, and perform corresponding control according to the selected transmission mode. Control unit: used to output control signals according to network status and adjust parameters for data packet generation and transmission. Work flow: Encrypt and obfuscate the encoded video stream; encapsulate the encrypted stream into data packets; select the transmission mode according to network conditions; send the data packet to the decoding end through the network; control the parameter transmission module 130 of the data packet according to the network status. The secondary encryption and encapsulation of the encoded stream and the selection of a suitable transmission mode realize optimal transmission control of the video stream, which is the guarantee that the video can be safely and efficiently distributed to the decoder through the public network. More specifically, the encryption unit applies encryption algorithms and technologies to re-encrypt the encoded

video stream to further improve the confidentiality and attack resistance of data transmission; the encapsulation unit uses packaging and encapsulation technology to generate data that can be transmitted on the network according to the transmission protocol. Packet and encapsulation processing shields the complexity of the underlying data transmission process; the transmission mode selection unit selects different transmission protocols and algorithms such as UDP, TCP or WebRTC based on real-time requirements, network bandwidth and other conditions to realize control of data packet transmission. Select the optimal mode; the transmission unit sends encapsulated data packets through the network interface based on the selected protocol, and handles transmission control and feedback; the control unit monitors the network status, dynamically adjusts and controls the parameters of data packet generation and transmission, and adapts to network changes ;Encryption and encapsulation improve data security. Transmission mode selection and transmission unit achieve efficient and reliable transmission. The control unit enables dynamic adjustment. The technical collaboration of each unit ensures reliable distribution of video streams; through secondary encryption, encapsulation technology and dynamic control of transmission modes and parameters, the transmission module 130 greatly improves the quality and efficiency of secure video transmission in the public network environment. To sum up, each unit in the transmission module 130 functions collaboratively through technology to achieve safe and reliable transmission of video streams, which is the core guarantee for video transmission. Specifically, the decoding module 140 is used to receive the video stream transmitted over the network, decode it, insert node information and embed watermark, and output the decoded video with watermark to the display terminal. Composed of units, the decoding module 140 includes a decoding unit, an information reading unit, an information insertion unit, a watermark embedding unit and an output unit. The decoding unit is used to decode the received video stream according to its own codec protocol and restore the video frame sequence. The information reading unit is used to parse the encoding node information in the video stream during the decoding process. An information insertion unit is used to insert decoding node information into the decoding frame sequence. The watermark embedding unit is used to select low-complexity frames to embed watermarks in decoded frames. The output unit is used to output decoded video frames with embedded node information and watermarks. The watermark embedding unit also includes a watermark generation subunit to generate a watermark containing a check code; a video analysis subunit to analyze the spatial and temporal information of each frame and evaluate the importance; an embedding control subunit to select the target frame based on importance; The watermark embedding subunit embeds the watermark in the target frame. The information insertion unit includes a coding information supplement subunit to supplement the complete encoding and decoding node information; an information steganography subunit to steganographically insert node information into the frame using digital watermark technology. Workflow: decode the received stream and restore the frame sequence; parse the encoding node information; insert the decoding node information in the decoding sequence; perform importance analysis on the decoded frame; embed the watermark in the low-complexity frame; output the decoding of

the embedded node and watermark While decoding the received video stream, the video decoding module 140 further inserts decoding node information and watermarks to generate complete node information required for video source traceability analysis. More specifically, the decoding unit uses the video decoding algorithm to decode the received stream according to the codec protocol, restore the video frame sequence, and realize the analysis and restoration of the compressed and encoded video; the information reading unit uses information steganalysis technology during the decoding process. It parses and extracts the node information inserted during encoding to provide basic data for traceability; the information insertion unit uses digital watermarking and other steganographic technologies to write decoding node information in the decoded frame to supplement the complete traceability node chain; the watermark embedding unit analyzes the video frame With watermark generation and embedding technology, hidden watermarks are implanted in decoded frames to enhance traceability reliability; the output unit combines processing of decoded frames and traceability information to output decoded videos with watermarks; the decoding unit implements video analysis. The information reading and insertion unit realizes traceability information collection. The watermark embedding unit enhances traceability reliability. The technologies of each unit work together to generate complete traceability information while decoding the video; through the organic cooperation of decoding, information insertion and watermark embedding technology, the decoding module 140 achieves reliable analysis of the compressed and encoded video and the integration of traceability information. In summary, each unit in the decoding module 140 jointly applies relevant technologies to ensure that both video decoding and traceability information extraction and generation are correct, providing a basis for subsequent traceability analysis. More specifically, the importance assessment of video frames needs to consider spatial domain information and temporal domain information; spatial domain information reflects the static picture content and can be analyzed through the following characteristics: Image complexity: reflects texture, edge and other detailed information, and can use gradient histogram represents; the number of targets, detecting the number of subjects in the picture through the target detection algorithm; the scene category, using scene recognition technology to determine the current scene category; the regional focus, analyzing the regional distribution of the image falling on the main target; the time domain information reflects dynamic changes, which can Analysis is performed through the following features: motion amount, the size of the motion vector between the current frame and the previous frame is calculated through the motion estimation algorithm; residual information, the inter-frame difference residual obtained by coding prediction; target motion, the detected target is between consecutive frames The position change; by fusing the above spatial and temporal features, a frame importance score for a specific application can be obtained. More specifically, image complexity reflects the detail level of the picture, mainly considering two aspects: spatial complexity and time complexity; spatial complexity can be judged by using the gradient information of the image: gradient extraction of the video frame to obtain the gradient amplitude Figure; calculate the standard deviation or variance of the gradient

amplitude map as an indicator of spatial complexity; set an appropriate threshold. When the standard deviation or variance is lower than the threshold, it is judged that the spatial complexity of the frame is low; the time complexity can use motion information and frame difference to judge: calculate the average of the motion vector sizes of the current frame and the previous frame as the amount of motion; calculate the mean square error between the current frame and the previous frame after motion compensation as the frame difference; if the amount of motion and the frame difference are both lower than the threshold, it is judged that the time complexity is low; the spatial complexity and the time complexity are combined, and when both are lower than the threshold at the same time, the image complexity of the frame is judged to be lower than the threshold; the setting of the threshold can be obtained through training and learning. Or adjust it for different applications.

Specifically, the display module is used to analyze, trace and display the video stream output by the decoding module 140, so as to realize traceability analysis of the video stream transmission process.

Component units: The display module includes a video display unit, a metadata analysis unit, a watermark analysis unit, a traceability analysis unit and a storage unit.

Video display unit: used to render and display the decoded video frame sequence on the display device.

Metadata analysis unit: used to analyze the encoding and decoding node time, location, device number and other information in the video frame.

Watermark analysis unit: used to detect and analyze watermark information in video frames.

Traceability analysis unit: used to fuse the analyzed node information to generate a complete video transmission trajectory.

Storage unit: used to store the generated video transmission trajectory information for source traceability query.

The traceability analysis unit includes:

- information association subunit: associates each node information according to timestamp, device number, etc.;
- route generation subunit: generates a complete transmission route based on the associated information.

Workflow: Render and display the decoded video on the display device; parse the encoding and decoding node information in the video frame; detect and parse the watermark in the video frame; associate and fuse the parsed node information; generate a complete transmission trajectory of the video; store the video transmission Trajectory, for traceability query and display module, by parsing and correlating the node information in the video, it can trace the source and generate the complete transmission link of the video, realizing traceable analysis of the video stream.

More specifically, the video display unit plays the decoded video frame sequence on the display device based on video encoding, decoding and rendering technology; the metadata analysis unit uses metadata analysis technology to extract and analyze the encoding and decoding node information in the video frame; watermark The analysis unit uses digital watermark analysis technology to detect and decode the watermark information in video frames; the traceability analysis unit uses information fusion and correlation technology to combine the analyzed node information to generate a complete transmission trajectory of the video; the storage unit is based on database storage technology. The generated video transmission track is stored for query; the display unit implements video playback. The parsing unit implements information extraction. The analysis unit realizes traceability generation. The storage unit implements data storage.

The technologies of each unit work together to achieve video analysis and traceability; through the organic application of video analysis, information extraction, correlation analysis and other technologies, the display module can trace the source to generate a complete transmission link of the video, achieving traceability analysis of the video content. In summary, each unit in the display module jointly applies relevant technical means to coordinate and cooperate to realize the analysis, traceability and storage functions of the video stream, forming a complete technical solution for video transmission traceability analysis. In summary, each module in this solution jointly completes the functions of information insertion, encoding, transmission, decoding, and analysis, which not only reduces the amount of data, but also provides various forms of key node information, which is achieved through mutual verification of information. Efficient and accurate video traceability. This collaborative working mechanism makes full use of the advantages of video technology and improves the efficiency of video traceability. Figure 2 is a schematic technical roadmap of a secure traceable video encoding and decoding system according to some embodiments of this specification. As shown in Figure 2, the video collection end collects the source video signal and obtains the source video stream in the source video stream. Inserting watermarks and metadata The watermark contains information such as time, location, device number, etc. The metadata contains content analysis information such as target detection and scene recognition. The video encoding end compresses and encodes the source video stream to improve network transmission efficiency. Insert encoding into the source video stream. Node information, including encoding time, location, device number, etc.; network video transmission uses protocols such as UDP, TCP or WebRTC to transmit video streams, and the appropriate transmission mode can be selected based on network conditions and real-time requirements; the video decoding end processes the received video stream Decode, restore the video frame sequence, insert decoding node information in the video frame, embed watermarks in the video frame, including encoding and decoding node information; the video display end displays and parses the video stream, generates a transmission trajectory, and extracts node information from the watermark and metadata. Perform traceability analysis and store the video transmission trajectory for subsequent traceability queries. Through the collaboration of video processing, transmission, and analysis, the source and circulation process of the video can be traced and monitored. Specifically, from the perspective of data flow direction, at the video collection end: source video signal → source video stream → source video stream after watermark insertion → generated metadata; at the video encoding end: source video stream → source video stream after insertion of encoding information Source video stream (after inserting encoding information) → Encoded and compressed video stream → Encapsulated video stream in network video transmission: Encapsulated video stream → Transmit data packet → Transmission module at the video decoding end: Transmit data packet → Decapsulation The decapsulated video stream → the decoded video frame sequence → the video frame sequence after inserting the decoded information the video frame sequence after inserting the decoded information → the video stream after embedding the watermark At the video display end: after embedding

the watermark Video stream → Video transmission track → Traceability analysis achieves a complete data flow from source video to transmission to display by inserting information into each processing link and parsing the information on the display end, effectively connecting the collection and processing of videos. , transmission, and analysis of each link, thereby enabling traceable analysis of video content. Figure 3 is an exemplary flow chart of a traceability analysis process according to some embodiments of this specification. As shown in Figure 3, the traceability analysis process includes: obtaining a decoded video stream; detecting and parsing node information in the video stream, including Encode node information and decode node information. Node information includes time, location, device number, etc.; detects watermark frames in the video stream and parses the node information in the watermark frame; associates and fuses the parsed encoding node information, decoding node information and node information in the watermark; according to The associated node information generates a complete transmission trajectory of the video stream; the generated video transmission trajectory is stored for subsequent source tracing queries. Figure 4 is an exemplary flow chart of a watermark embedding process according to some embodiments of this specification. As shown in Figure 4, the watermark embedding process includes: generating a watermark containing a check code according to node information; analyzing the spatial and temporal domains of the video frame. Domain information, evaluate the importance of each frame; according to the importance threshold, select frames with low importance as watermarks to embed in the target frame; embed the generated watermark containing node information in the target frame. Video frames of low importance can be frames whose image complexity is lower than the threshold. The embedding process needs to control the intensity of the watermark to ensure video quality. The embedding position also needs to be optimized to ensure the watermark detection rate. The watermark needs to add a verification mechanism to detect the watermark. For completeness, the embedding parameters need to be dynamically adjusted according to the video content. The invention and its implementation have been schematically described above. This description is not limiting. The invention can be implemented in other specific forms without departing from the spirit or basic characteristics of the invention. What is shown in the drawings is only one embodiment of the present invention, and the actual structure is not limited thereto. Any reference signs in the claims shall not limit the claims involved. Therefore, if a person of ordinary skill in the art is inspired by the invention and without deviating from the purpose of the invention, can creatively design structural methods and embodiments similar to the technical solution, they shall all fall within the protection scope of this patent. Furthermore, the word "comprising" does not exclude other elements or steps, and the word "a" before an element does not exclude the inclusion of "a plurality" of that element. Multiple elements stated in a product claim may also be implemented by one element through software or hardware. Words such as first and second are used to indicate names and do not indicate any specific order. Claims (English machine translation) Global Dossier EN ZH Data originating from sources other than the EPO may not be accurate, complete, or up to date. The wording below is an initial machine translation of the original

publication. To generate a version using the latest translation technology, go to the original language text and use Patent Translate.

1. A secure and traceable video encoding and decoding system, including: The video collection module is used to collect the source video stream and insert watermark information and metadata into the source video stream; A video encoding module, used to receive the source video stream, insert encoding node information into the source video stream, and compress and encode the video stream after inserting the encoding node information; The transmission module is used to encapsulate and transmit the encoded video stream output by the video encoding module; The decoding module is used to receive the video stream output by the transmission module, decode the received video stream, and insert decoding node information and watermarks into the decoded video stream; The display module is used to display the video stream output by the decoding module, parse node information and watermarks from it, and generate the transmission trajectory of the video stream; Among them, the display module includes a traceability analysis unit, which is used to obtain the decoded video stream, analyze the node information and watermarks in the video stream, and track the image propagation link.

2. The safe and traceable video encoding and decoding system according to claim 1, characterized in that: Video capture module includes: A video source unit used to obtain the source video signal generated by the video output device; A coding standard selection unit, used to select the corresponding video coding standard according to the format and system requirements of the source video signal. The video coding standard includes H.264 and H.265; A video acquisition unit, used to collect source video signals according to the selected video coding standard and generate a source video stream that meets the requirements of the video coding standard; The image enhancement unit is used to perform image quality enhancement processing on the source video stream. The image enhancement processing includes denoising and color correction; A watermark information unit, used to insert watermark information at a preset position of the source video stream, where the watermark information includes video collection time, location and device number information; A metadata generation unit, used to extract frame images in the source video stream as key frames at preset intervals, and generate metadata describing the video content by analyzing the key frames. The metadata includes target detection and scene recognition. information; A collection control unit, used to control the working mode of the video collection module according to the network status and user instructions; Wherein, the video output device includes a camera and a computer operating system desktop.

3. The safe and traceable video encoding and decoding system according to claim 1, characterized in that: Video encoding modules include: An information insertion unit, configured to receive the source video stream, insert node information into the source video stream, and generate a video frame sequence with node information, where the node information includes encoding time, location and encoder number information; Coding unit is used to receive the video frame sequence with node information, select the H.264 or H.265 video coding standard according to the network status, adjust the code rate control parameters and variable block size parameters of the encoder, and compress the video frame sequence Encoding, output the

compressed and encoded video frame sequence; An encryption unit, configured to receive a compressed video frame sequence, encrypt the video frame sequence using a symmetric or asymmetric encryption algorithm, and output the encrypted video frame sequence; The encapsulation unit is used to receive the encrypted video frame sequence and encapsulate it into an output code stream that complies with the video transmission protocol standard. 4. The safe and traceable video encoding and decoding system according to claim 1, characterized in that: Transport modules include: The encryption unit is used to receive the encoded video frame sequence output by the video encoding module, obfuscate it using an encryption algorithm, and output the obfuscated video frame sequence; The encapsulation unit is connected to the encryption unit and is used to encapsulate the obfuscated video frame sequence into transmission data packets; A transmission mode selection unit, used to select a transmission mode and output a control signal according to the real-time requirements, network bandwidth and stability of the video stream. The transmission mode includes UDP, TCP and WebRTC; The transmission unit is used to receive transmission data packets and send the transmission data packets to the video decoding module through the network according to the control signal of the transmission mode selection unit; The control unit is used to output control signals according to the network status and adjust the generation and transmission parameters of data packets. 5. The safe and traceable video encoding and decoding system according to claim 1, characterized in that: Decoding modules include: The decoding unit is used to receive the video stream output by the video transmission module, decode the received video stream according to its own codec protocol, and restore the video frame sequence; The information reading unit is used to read and parse the node information in the video frame sequence when decoding it; An information insertion unit is used to receive the decoded video frame sequence and insert decoding node information into the video frame sequence; The watermark embedding unit is used to select low-visibility video frames from the decoded and restored video frames and embed watermarks containing information about each node; The output unit is used to output a sequence of video frames embedding decoding node information and watermarks. 6. The safe and traceable video encoding and decoding system according to claim 5, characterized in that: The watermark embedding unit includes: The watermark generation subunit is used to generate a watermark containing a check code based on node information; The video analysis subunit is used to analyze the spatial and temporal information of video frames and obtain the importance of each video frame; The embedding control subunit is used to select the video frame whose importance is lower than the threshold as the target frame for watermark embedding according to the importance of each acquired video frame; Watermark embedding subunit, used to embed the generated watermark in the target frame; Wherein, the video frames whose importance is lower than the threshold are video frames whose image complexity is lower than the threshold. 7. The safe and traceable video encoding and decoding system according to claim 5, characterized in that: Information insertion units include: The coding information supplement subunit is used to supplement the complete coding and decoding node information according to the decoding time and device number;

The information steganography subunit is used to use steganography technology to extract and insert the supplemented node information into the video frame. 8. The safe and traceable video encoding and decoding system according to claim 7, characterized in that: Steganography technology is digital watermarking technology. 9. The safe and traceable video encoding and decoding system according to claim 1, characterized in that: Display modules include: A video display unit, used to receive the decoded video frame sequence and render and display it on the display device; The metadata parsing unit is used to detect and parse the information frames in the video frame sequence, and obtain the time, location and device number information of each encoding and decoding node contained therein; The watermark analysis unit is used to detect watermark frames in the video frame sequence and analyze the node information watermarks in them; The traceability analysis unit is used to fuse the node information obtained by the metadata analysis unit and the watermark analysis unit to form a complete video transmission trajectory from the source to the destination node; The storage unit is used to store video transmission trajectory information for traceability and query. 10. The secure and traceable video encoding and decoding system according to claim 9, characterized in that: Traceability analysis units include: The information correlation subunit is used to correlate the parsed node information based on the timestamp and device number information; The route generation subunit is used to generate a complete video stream transmission route based on the associated node information.